

Deepak Mallampati

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Kent, OH, USA -- Remote / international teams (F-1 OPT)

PROFESSIONAL SUMMARY

AI Automation Engineer (2.5+ years) who designs and ships **LLM-driven automation workflows** end-to-end — integrating models (Claude, GPT-4o, Gemini) and external APIs into reliable, cost-efficient business pipelines. Built fault-tolerant **automation infrastructure** with scheduled orchestration and exponential-backoff retries that cut failed production runs **~60%**, and operated **25+ production AI agents** serving **100K+ requests/day** across 20+ teams. Fluent in **Python and JavaScript/TypeScript**, REST APIs, webhooks, and database integrations; ships backend logic and inline tools (Manga Lens, Chrome Web Store) from problem to production in days. Comfortable owning workflow reliability, latency, and per-query cost, and translating cross-functional operational friction into shipped automations.

CORE COMPETENCIES

LLM-Driven Workflow Automation . AI / LLM API Integration (OpenAI, Anthropic, HF) . Fault-Tolerant Automation Pipelines . REST APIs / Webhooks / DB Integration . Python & JavaScript / TypeScript . Agentic LLM Systems . Production RAG & Evaluation . Performance, Reliability & Cost Optimization . Cross-Functional Operational Delivery . End-to-End Ownership

EXPERIENCE

Progress Solutions Inc.

Jul 2025 -- Present

Jr. AI/ML Engineer Trainee -- USA | Healthcare Technology, AI Consulting

- Built **agentic LLM workflows** for multi-step healthcare queries (eligibility checks, care navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; improved agent stability **~25%** and reduced hallucination rate **>30%**.
- Built **RAG systems** for clinical knowledge retrieval; recursive semantic chunking + transformer embeddings improved precision **~35%** and reduced irrelevant retrieval **>30%**; response alignment exceeded **90%**.
- Packaged LLM/ML inference as **FastAPI/Flask** RESTful services containerized with Docker; structured logging + load simulation cut post-deployment defects **~30%**.
- Authored stakeholder-facing system-limitation docs and audit trails for non-technical clinical operators — enabled adoption in HIPAA-conscious environment.
- Shipped predictive ML (scikit-learn, XGBoost) for patient no-show and care engagement; improved recall **15-20%** on high-risk while holding precision **>90%**.

Energy Solutions International (Emerson)

Jun 2022 -- Apr 2023

Database & DevOps Performance Engineer (Intern) -- Hyderabad, India | Enterprise ERP

- Built CI/CD pipelines with Jenkins for the Synthesis Order-to-Cash ERP platform; reduced deployment errors **>30%**, improved release cycle **~35-40%**.
- Designed Grafana observability dashboards for production SQL systems; cut incident recurrence **~25%**, improved RCA speed **~30%**.

Suvidha Foundation

Jan 2022 -- Mar 2022

Machine Learning Engineer -- Hyderabad, India | Nonprofit Tech, Applied NLP

- Built transformer-based video summarization for 5,000+ sessions; reduced manual review time **60-70%**; deployed as Flask API for non-technical users.
- Implemented document Q&A with RAG; reduced hallucinated outputs **~30%**.

PROJECTS

Manga Lens shipped -- Chrome Web Store

Solo-built Chrome extension integrating **4 AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, local Ollama) via provider-agnostic abstraction. Multi-section capture pipeline for tall webtoon panels, 7-day translation cache, 29 site selector configs. Full ship discipline: Manifest V3, service workers, scoped permissions, privacy policy, Chrome Web Store review. **Demonstrates fast solo ship from problem to production.**

TypeScript, Manifest V3, content scripts, multi-provider abstraction, captureVisibleTab

Agentic Healthcare Claims Processing & Fraud Risk Intelligence

Multi-agent pipeline (Intake -> Validation -> Consistency -> Duplicate Detection -> Fraud Risk) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation; ANN similarity for duplicate detection; explainable risk scoring with audit-ready reasoning traces. Direct analog to insurance underwriting / brokerage operational AI workflows.

Python, LangChain, vector search, FastAPI, Docker, regulated-domain governance

Dream Decoder -- AI Multimodal Platform

Full-stack FastAPI + React/TypeScript app coordinating multimodal AI APIs (Perplexity Sonar for interpretation, Replicate for visuals). Intermediate structured prompt transformation layers improved contextual alignment **~30%** and first-pass image success **~25-30%**.

FastAPI, React, TypeScript, Vite, multi-provider LLM orchestration

EDUCATION

Master's Degree -- Kent State University, Ohio, USA

Aug 2023 -- May 2025

TECHNICAL SKILLS

Languages & Frameworks: Python, TypeScript, JavaScript, React, FastAPI, Flask, SQL (PostgreSQL, T-SQL, SQLite)

AI/LLM & Agentic: LLM APIs (Claude, GPT-4o, Gemini, Ollama), LangChain, LlamaIndex, RAG, multi-agent workflows, schema contracts, prompt engineering, structured outputs, evaluation pipelines, guardrails, grounding

Internal Tools / Automation: Chrome extensions (Manifest V3), full-stack web apps (FastAPI + React), workflow orchestration, multi-provider API integration

Developer Tooling: Cursor, Claude Code, Docker, CI/CD (Jenkins), Grafana, REST APIs, MLOps/LLMOps, observability/logging

Regulated Domains: HIPAA-conscious governance, de-identification, audit trails, stakeholder-facing system-limitation docs — transferable to CFPB / state insurance compliance

Work Authorization: F-1 OPT -- authorized to work in the United States now; future sponsorship required